



Annuities Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 Ella contributes \$1200 per month at the start of each month into an investment which pays interest at 3.6 % per annum, compounded monthly.

- a. Show that at the end of the second month, the future value of Ella's investment will be 2

\$2410.81.

$r = 3.6\% \text{ pa}$
 $= \frac{3.6}{12} \% \text{ per month}$
 $= 0.3\% = 0.003$

At end of 2nd month $FV = 1200(1.003)^2 + 1200(1.003)$
 $= \$2410.8108 = \2410.81

- b. After 18 months, the future value of Ella's investment will be \$22 226.19. Beth has a lump sum of \$21 600 which is equivalent to the total of Ella's contributions. Beth invests her lump sum into the same investment account paying 3.6% per annum, compounded monthly. How much more than Ella will Beth have in her investment account after 18 months?

$P = \$21600$ $r = 1.003$ $n = 18$

Beth's account: $FV = 21600(1.003)^{18}$
 $= \$22796.62$ ✓

difference between FV for Ella and Beth
 $= 22796.62 - 22226.19$
 $= \$570.43$ ✓

∴ Beth will have \$570.43 more in her account after 18 months

- 2 Present value interest factors for an annuity of \$1 for various interest rates and time periods are given in the table below. Isaac uses this table to calculate his car loan repayments. His loan is \$24 500 and will be paid in equal monthly payments. The

Table of present value interest factors

Time periods	Interest rate per period			
	0.4%	0.5%	0.6%	0.7%
12	11.6937	11.6189	11.5448	11.4714
24	22.8405	22.5629	22.2899	22.0216
36	33.4659	32.8710	32.2907	31.7247
48	43.5942	42.5803	41.5988	40.6486

monthly interest rate for his loan is 0.7%. Calculate the how much Isaac will save paying back the loan in 2 years instead of 3.

Monthly instalments for 2 years: $24500 \div 22.0216 = \$1112.54$
Total instalments for 2 years: $1112.54 \times 24 = \$26701.06$
Monthly instalments for 3 years: $24500 \div 31.7247 = \$772.27$
Total instalments for 3 years: $772.27 \times 36 = \$27801.68$
Savings = $27801.68 - 26701.06 = \$1100.62$

- 3 Alexa is buying her first car. She takes out a loan for \$15 000 and agrees to repay the loan in monthly repayments at 7.2% per annum for 5 years. Use the present value of an annuity table to determine how much interest she will pay on the loan.

3

Table of present value interest factors						
N \ r	Interest rate per period					
	0.5%	0.6%	0.7%	0.8%	0.9%	1%
36	32.8710	32.2907	31.7247	31.1723	30.6334	30.1075
48	42.5803	41.5988	40.6486	39.7284	38.8372	37.9740
60	51.7256	50.2621	48.8559	47.5042	46.2047	44.9550
72	60.3395	58.3253	56.4041	54.5710	52.8212	51.1504

$$\begin{aligned}
 r &= 7.2\% \text{ pa} & n &= 5 \text{ years} \\
 &= 7.2 \div 12 \% \text{ per month} & &= 5 \times 12 \text{ months} \\
 &= 0.6\% & &= 60 \\
 \text{table value} &= 50.2621 \quad \checkmark \\
 PV &= a \times \text{table value} \\
 15000 &= a \times 50.2621 \\
 \frac{15000}{50.2621} &= a \\
 a &= 298.4356... \\
 &= \$298.44 \quad \checkmark \\
 \text{interest} &= (\$298.44 \times 60) - \$15000 \\
 &= 17906.40 - 15000 \\
 &= \$2906.40 \quad \checkmark
 \end{aligned}$$

- 4 Tom has a total of \$80 000 in an investment account earning interest at a rate of 0.6% per month. Each month, immediately after the interest has been paid, Tom withdraws \$1200 from the account. The amount in the account immediately after the n th withdrawal can be determined using the recurrence relation

$$A_n = A_{n-1}(1.006) - 1200$$

$$\text{where } n = 1, 2, 3, \dots \text{ and } A_0 = 80\,000$$

- a. Use the recurrence relation to find the amount of money in the account immediately after the third withdrawal.

2

$$\begin{aligned}
 A_0 &= \$80000 \\
 A_1 &= 80000(1.006) - 1200 \\
 &= \$79280 \\
 A_2 &= 79280(1.006) - 1200 \\
 &= \$78555.68 \\
 A_3 &= 78555.68(1.006) - 1200 \\
 &= \$77827.01408 \\
 &= \$77827.01 \quad \checkmark
 \end{aligned}$$

- b. Calculate the amount of interest earned in the first three months.

2

$$\begin{aligned}
 \text{interest} &= 77827.01 + (1200 \times 3) - 80000 \\
 &= 81427.01 - 80000 \\
 &= \$1427.01 \quad \checkmark
 \end{aligned}$$

- 5 Adam makes an initial deposit of \$5000 on an investment taken out over one year at a rate of 7.2% per annum compounded quarterly, and an additional deposit of \$200 is made each quarter. The amount in the account immediately after the n th deposit can be determined using the recurrence relation

$$A_{n+1} = A_n(1 + r) + 200$$

where $n = 1, 2, 3, \dots$ and $A_0 = 5\,000$

A_{n+1} is the value of the loan after $(n + 1)$ repayments

A_n is the value of the loan after n repayments

r is the rate of interest

- a. Find the value of r correct to two significant figures. $r = 0.072 \div 4 = 0.018$ 1
- b. Use the recurrence relation to find the amount of money in the account immediately after the second deposit. 2

$$5000(1 + 0.018) + 200 = 5290$$

$$5290(1 + 0.018) + 200 = 5585.22$$

Therefore \$5585.22

- 6 The table below shows the future value of a \$1 annuity.

Future value of \$1						
Period	1%	2%	3%	4%	5%	6%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500	2.0600
3	3.0301	3.0604	3.0909	3.1216	3.1525	3.1836
4	4.0604	4.1216	4.1836	4.2465	4.3101	4.3746

- a. An annuity of \$14 800 is invested every half-year at 6% per annum, compounded six-monthly for 2 years. What is the future value of the annuity? 1

Intersection value is 4.1836 (3% and 4 years)

$$FV = 4.1836 \times 14\,800 = \$61917.28$$

- b. Sonia aims to have a deposit for an apartment of at least \$180 000 in 3 years-time by investing in an annuity. The annuity has an interest rate of 2% p.a. compounded annually. Calculate Sonia's yearly contribution to achieve the deposit. Answer to the nearest dollar. 2

Intersection value is 3.0604 (2% and 3 years)

Let x be the amount to be invested

$$FV = 3.0604 \times x$$

$$180\,000 = 3.0604 \times x$$

$$x = \frac{180\,000}{3.0604} = \$58815.84107 \dots \approx \$58816$$

\$58816 is invested every year.

- 7 The table of interest factors below shows the future value of a \$1 annuity for different interest rates over different numbers of time periods.

Future value of an annuity of \$1

Time periods	Interest rate per period					
	1%	2%	3%	4%	5%	6%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500	2.0600
3	3.0301	3.0604	3.0909	3.1216	3.1525	3.1836
4	4.0604	4.1216	4.1836	4.2465	4.3101	4.3746

- a. Colleen makes equal contributions of \$5500 into an account every 6 months for 2 years, at an interest rate of 4% per annum compounding biannually. What is the future value of Colleen's annuity? 1

$$FV = 5500 \times 4.1216 = \$22,668.80$$

- b. Claire wants to make 4 equal contributions throughout the year into an account with an interest rate of 4% per annum, compounding quarterly. These contributions will be whole dollar amounts. She aims to have \$12 000 in the account by the end of the year. Calculate the least amount her contributions need to be. 2

$$\text{Instalment} \times 4.0604 = 12000$$

$$\text{Instalment} = 12000 \div 4.0604 = \$2955.37 \dots$$

So the instalments need to be \$2956

- 8 Jacob makes an initial deposit of \$ 70000 on an investment, earning an interest rate of 0.5% per month. An initial amount of \$800 is made every month. The amount in the account can be determined by using the recurrence relation

$$A_n = A_{n-1}(1.005) + 800$$

- a. Use the recurrence relation to find the amount of money in the account after the third deposit. 2

$$\begin{aligned} A_1 &= 70000(1.005) + 800 = 71150 \\ A_2 &= 71150(1.005) + 800 = 72305.75 \\ A_3 &= 72305.75(1.005) + 800 = 73467.2788 \end{aligned}$$

- b. Calculate the amount of interest earned in the first three months. 2

$$\begin{aligned} \text{Amount} &= 70000 + (800 \times 3) = 72400 \\ \text{Interest} &= 73467.2788 - 72400 \\ &= 1067.2788 \end{aligned}$$

- 9 A table of future value interest factors for an annuity of \$1 is shown below

Period	2%	2.50%	3%	3.50%	4%	4.50%	5%	5.50%
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2	2.020	2.025	2.030	2.035	2.040	2.045	2.050	2.055
3	3.060	3.075	3.090	3.106	3.121	3.137	3.152	3.168
4	4.121	4.152	4.183	4.214	4.246	4.278	4.310	4.342
5	5.204	5.256	5.309	5.362	5.416	5.470	5.525	5.581
6	6.308	6.387	6.468	6.550	6.633	6.716	6.801	6.888
7	7.434	7.547	7.662	7.779	7.898	8.019	8.142	8.266
8	8.583	8.736	8.892	9.051	9.214	9.380	9.549	9.721
9	9.754	9.954	10.159	10.368	10.582	10.802	11.026	11.256

- a. Find the future value of an annuity with a contribution of \$1500 per quarter for 24 months at 8% per annum compounded quarterly. 1

$$1500 \times 8.583 = 12874.5$$

- b. Calculate the interest earned on this annuity. 1

$$12874.5 - 1500 \times 8 = 874.5$$

- c. Find the contribution to be made at the end of each year, for 4 years that will give a future value of \$4500 if the annuity earns 4.5% compounded annually. 1

$$\begin{aligned}
 4.278 \times \text{Contribution} &= 4500 \\
 \text{Contribution} &= \frac{4500}{4.278} \\
 &= 1051.893
 \end{aligned}$$



Annuities Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 The table below shows the future value of a \$1 annuity at different interest rates over different numbers of time periods. Patrick invests \$300 bi-annually at an interest rate of 4% pa. How much interest will he earn after 4 years? 1

Period	Interest rate per period				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

A. \$174.90

- 2 To purchase her first car, Tamara obtains a loan of \$9500. The loan is borrowed at 7.2% per annum, compounded quarterly. Each quarter, after the interest has been applied, Tamara makes a repayment of \$450. The recurrence relation is modelled by $V_{n+1} = V_n \times (1 + r) - 450$,

- V_{n+1} is the value of the loan after $(n + 1)$ repayments
- V_n is the value of the loan after n repayments
- r is the rate of interest.

a. Find the value of r correct to two significant figures. $r = \frac{7.2\%}{4} = 0.018$ OR 1.8% 1

b. Use the recurrence relation to calculate the balance of the loan after Tamara has made her second repayment. 2

$$V_{n+1} = V_n \times (1 + r) - 450$$

After the first repayment:

$$V_{n+1} = 9500 \times (1 + 0.018) - 450 = \$9221$$

After the second repayment:

$$V_{n+1} = 9221 \times (1 + 0.018) - 450 = \$8936.98$$

- 3 Hanako contributes \$8500 to an investment account every year for 6 years. The account pays 8% interest per annum, compounded annually.

Future value of \$1								
Period	1%	2%	3%	4%	5%	6%	7%	8%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500	2.0600	2.0700	2.0800
3	3.0301	3.0604	3.0909	3.1216	3.1525	3.1836	3.2149	3.2464
4	4.0604	4.1216	4.1836	4.2465	4.3101	4.3746	4.4399	4.5061
5	5.1010	5.2040	5.3091	5.4163	5.5256	5.6371	5.7507	5.8666
6	6.1520	6.3081	6.4684	6.6330	6.8019	6.9753	7.1533	7.3359
7	7.2135	7.4343	7.6625	7.8983	8.1420	8.3938	8.6540	8.9228
8	8.2857	8.5380	8.8923	9.2142	9.5491	9.8975	10.2598	10.6366

- a. What is the future value of Hanako's annuity at the end of 6 years? 1

$$FV = 8500 \times 7.3359 = \$62355.15$$

- b. Calculate the total amount of interest that Hanako will earn on her investment at the end of 6 years. 1

$$\text{total amount contributed} = 8500 \times 6 = \$51000$$

$$\text{total amount of interest} = \text{future value of the annuity}$$

$$- \text{total amount contributed}$$

$$= 62355.15 - 51000 = \$11355.15$$

- c. Hanako is considering opening another investment account that pays 8% interest per annum, compounded biannually, for 3 years. Using the formula $FV = PV(1 + r)^n$, 2

determine the single sum of money that Hanako would need to invest in this account to reach the same future value calculated in part (a).

$$FV = PV(1 + r)^n$$

$$62355.15 = PV \left(1 + \frac{8\%}{2} \right)^{3 \times 2}$$

$$PV = \frac{62355.15}{\left(1 + \frac{8\%}{2} \right)^6} = \$49280.18$$

- 4 The table below gives the present value of an annuity of \$1 at the 3

given interest rate for the given period. Sharon takes out a loan of \$12000 to buy a car. This loan is to be repaid half yearly over 4 years

Present Value Interest Factors (PVA)								
\$1	Interest Rate Per Period							
N	1%	2%	3%	4%	5%	6%	7%	8%
1	0.9901	0.9804	0.9709	0.9615	0.9524	0.9434	0.9346	0.9269
2	1.9704	1.9416	1.9135	1.8861	1.8594	1.8334	1.8080	1.7833
3	2.9410	2.8839	2.8286	2.7751	2.7232	2.6730	2.6243	2.5771
4	3.9020	3.8077	3.7171	3.6299	3.5460	3.4651	3.3872	3.3121
5	4.8545	4.7135	4.5797	4.4518	4.3295	4.2124	4.1002	3.9927
6	5.7955	5.6014	5.4172	5.2421	5.0757	4.9173	4.7665	4.6229
7	6.7282	6.4720	6.2303	6.0021	5.7864	5.5824	5.3893	5.2064
8	7.6517	7.3255	7.0197	6.7327	6.4632	6.2098	5.9713	5.7466
9	8.5660	8.1622	7.7861	7.4353	7.1078	6.8017	6.5152	6.2469
10	9.4713	8.9826	8.5302	8.1109	7.7217	7.3601	7.0236	6.7101

at an interest rate of 8% per year. By first using the Present Value Interest Factor table to find his half yearly repayments, calculate the amount of interest Sharon will pay over the life of his loan.

$$12000 \div 6.7327 = 1782.34 \dots$$

$$\text{total repayed} = 1782.34 \times 4 \times 2 = 14258.76$$

$$\text{Interest} = 14258.76 - 12000 = \$2258.77$$

- 5 The table gives the present value interest factors for an annuity of \$1 per period. For various interest rates (R) and number of periods (N).

Table of present value interest factors					
r N	Interest rate per period (as a decimal)				
	0.0075	0.0080	0.0085	0.0090	0.0095
70	54.30462	53.43960	52.59397	51.76724	50.95891
71	54.89293	54.00754	53.14226	52.29657	51.46995
72	55.47685	54.57097	53.68593	52.82118	51.97618
73	56.05643	55.12993	54.22502	53.34111	52.47764
74	56.63169	55.68446	54.75957	53.85641	52.97438

- a. Oscar plans to invest \$200 each month for 74 months. His investment will earn interest at the rate of 0.0080 (as a decimal) per month. Use this information in the table to calculate the present value of this annuity. 1

$$PV = 200 \times 55.68446 = 11136.89$$

- b. Lucy is using the same table to calculate the loan repayment for her car loan. Her loan is \$21 500 and will be repaid in equal monthly repayments over 6 years. The interest rate on her loan is 10.8% per annum. Calculate the amount of each monthly repayment, correct to the nearest dollar. 2

$$0.108 \div 12 = 0.0090$$

repayment is the annuity

$$21500 = 52.82118 \times \text{repayment}$$

$$\text{repayment} = \$407.03 = \$407 \checkmark$$

- 6 Connor and Lani would like to buy a property and have been saving for their deposit.

- a. The average appreciation for property in Australia over the past number of years is 7% p.a. compounding annually. If the property market continues to increase at this rate, how much will a property currently valued at \$800 000 be worth in 3 years? 1

$$\$800\,000 (1 + 0.07)^3 = \$980\,034.40$$

- b. If they can save a 20% deposit, they will not have to pay lender's mortgage insurance. They have \$80 000 in savings now. If they save \$3 400 per month over the next three years, will they have a large enough deposit to avoid paying the lender's mortgage insurance for a mortgage on a property of this value at the end of 3 years? Fully justify your answer with all relevant calculations. 2

Deposit required:

$$20\% \times \$980\,034.40 = \$196\,006.88$$

Savings:

$$\$80\,000 + (\$3\,400 \times 12 \times 3) = \$202\,400$$

Yes. They will have saved \$6 393.12 more than 20% of the projected property price so they will avoid paying lenders mortgage insurance.

- c. By the end of the three years, they decide to take out a mortgage of \$900,000 to purchase their property over 30 years at 3% p.a. compounding monthly. Use the present value table to calculate the amount of their monthly mortgage repayment. 3

Present value of an annuity of \$1,000

Period	Interest rate per period (%)					
	0.20	0.25	0.30	0.35	0.40	0.45
120	0.107	0.104	0.101	0.098	0.095	0.093
240	0.190	0.180	0.171	0.162	0.154	0.147
360	0.256	0.237	0.220	0.204	0.191	0.178
400	0.275	0.253	0.233	0.215	0.199	0.185
600	0.349	0.311	0.278	0.251	0.227	0.207

Rate: 3% p.a. $\div 12 = 0.25\%$ per month

Period: 3×12 months = 36 months

$\$900\,000 \div 1000 = 900$

From the present value table:

$900 \div 0.237 = 3797.4683 \dots$

Monthly repayment = \$3 797.47

- d. Calculate the amount of interest paid on this mortgage. 2

Total paid = $\$3\,797.47 \times 12 \times 30 = \$1\,367\,089.20$

Interest = $\$1\,367\,089.20 - \$900\,000 = \$467\,089.20$

- 7 a. Scott would like to save some money for his niece starting from when she is born. He gets paid weekly and could set aside \$50 per week. If he gives her the money when she turns eighteen, how much would she get? 1

$\$50 \times 52 \times 18 = \$46\,800$

- b. Scott decides to begin an annuity for her instead. He deposits \$2 500 when she is born and at each birthday up to her 18th birthday. On her 18th birthday he will withdraw the money. If the interest rate is 0.5% p.a., use the table below to calculate the value of the annuity when his niece turns 18. 2

Future value of \$1 invested

Period	Interest rate per period (%)					
	0.30	0.35	0.40	0.45	0.50	0.55
15	15.3191	15.3731	15.4274	15.4818	15.5365	15.5915
18	18.4664	18.5456	18.6253	18.7053	18.7858	18.8667
21	21.6421	21.7516	21.8617	21.9725	22.0840	22.1962
24	24.8465	24.9913	25.1371	25.2840	25.4320	25.5810
27	28.0798	28.2651	28.4519	28.6404	28.8304	29.0220
30	31.3423	31.5734	31.8068	32.0423	32.2800	32.5200

From the table, 0.5% interest rate for 18 periods = 18.7858

$\$2\,500 \times 18.7858 = \$46\,964.50$

- c. Explain, with calculations, which is the better option for both Scott and his niece. 2

The annuity is a better option for both Scott and his niece.

Niece:

$\$46\,964.50 - \$46\,800 = \$164.50$ extra for her when she is 18.

Scott:

$\$2\,500 \times 18 \text{ years} = \$45\,000$

OR

$\$50 \times 52 \text{ weeks} = \$2\,600$

$\$46\,800 - \$45\,000 = \$1\,800$ total savings.

$\$2\,600 - \$2\,500 = \$100$ saved per year.

8 Wilma deposited a lump sum into a new bank account which earns 2% per annum

3

compound interest. Present value interest factors for an annuity of \$1 for various interest rates (r) and numbers of periods (N) are given in the table.

Table of present value interest factors

$\begin{matrix} r \\ N \end{matrix}$	<i>Interest rate per period as a decimal</i>			
	0.01	0.015	0.02	0.025
10	9.471	9.222	8.983	8.752
20	18.046	17.169	16.351	15.589
30	25.808	24.016	22.396	20.930

Wilma was able to make the following withdrawals from this account.

- \$1000 at the end of each year for twenty years (starting one year after the account is opened)
- \$3000 each year for ten years starting 21 years after the account is opened.

Calculate the minimum lump sum Wilma must have deposited when she opened the new account.

For the first 20 years she needs $PV = \$1000 \times 16.351 = \16352 .

For the next 10 years she needs $PV = \$3000 \times 8.983 = \26949 at the start of the 21st year.

At the start, i.e. 20 years earlier, \$26949 is considered as the future value FV .

$$\therefore \text{the present value } PV \text{ (i.e. value at the start)} = \frac{26949}{1.02^{20}} \approx \$18135.90$$

$$\therefore \text{minimum opening lump sum} = \$16352 + \$18135.90 = \$34486.90$$